7.3 Determinants

To every square matrix $A = [a_{ij}]$ of order n, we can associate a number called **determinant** of the matrix A.

If $A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix}$, then determinant of A is written as $|A| = \begin{vmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{vmatrix}$.

Note 7.5

(i) Determinants can be defined only for square matrices.

(ii) For a square matrix A, $|A|$ is read as **determinant of *A*.**

(iii) Matrix is only a representation whereas determinant is a value of a matrix.

7.3.1 Determinants of Matrices of different order

Determinant of a matrix of order 1

Let $A = [a]$ be the matrix of order 1, then the determinant of A is defined as 'a'.

Determinant of a matrix of order 2

Let $A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$ be a matrix of order 2. Then the determinant of A is defined as

$$|A| = \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = a_{11}\ a_{22} - a_{21}\ a_{12}.$$

Example 7.14

Evaluate : (i) $\begin{vmatrix} 2 & 4 \\ -1 & 2 \end{vmatrix}$ (ii) $\begin{vmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{vmatrix}$.

Solution

(i) $\begin{vmatrix} 2 & 4 \\ -1 & 2 \end{vmatrix} = (2 \times 2) - (-1 \times 4) = 4 + 4 = 8.$

(ii) $\begin{vmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{vmatrix} = (\cos\theta\cos\theta) - (-\sin\theta\sin\theta) = \cos^2\theta + \sin^2\theta = 1.$

Determinant of a Matrix of order 3

We consider the determinant of a 3 × 3 matrix with entries as real numbers or real valued functions defined on $\mathbb{R}$ and study its properties and discuss various methods of evaluation of certain determinants.

Definition 7.19

Let $A = [a_{ij}]_{3\times3}$ be a given square matrix of order 3. The **minor** of an arbitrary element a_{ij} is the determinant obtained by deleting the ith row and j^{th} column in which the element a_{ij} stands. The minor of a_{ij} is usually denoted by M_{ij}.

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